Digital marketing: This field has revolutionised modern businesses. While the amount of information on potential consumers grows, AI-related technology will be of utmost importance when making data-based decisions. AI helps find people and customers based on their interests, demographics and other aspects to learn and detect the best audience for particular brands.

Content creation: There are areas where content created by AI can be useful and help attract visitors to a website. AI can also write reports and news based on data and information. Hundreds of articles can be created with AI technology quickly, which can save a lot of time and resources.

Online searches: Old ways of performing online searches no longer stay true. Two use cases using AI that have revolutionised Internet searches and search engine optimisation (SEO) are voice search and Google's algorithm called RankBrain. These have changed the way marketeers create and optimise their Web content.

Web designing: With AI, websites could exist without the help of programmers and designers. Applications, such as Grid, use AI to design websites based on the information provided by users like images, text, calls-to-action, etc. AI can make websites look professional in very little time and at a much lower cost.

Chatbots: Consumers are already using chatbots to chat with friends and colleagues without waiting for a long time for a response. Chatbots automate responses to potential buyers' frequently asked questions and provide them a way to search for the product or service they are looking for.

Natural learning processing and machine learning techniques are used by these bots to find the correct responses. Many brands have started using these techniques to communicate with their prospective customers through messenger applications like Facebook Messenger, WhatsApp and Slack.

Cybersecurity: The main concern in today's digital world is cybersecurity. Malware and virus attacks are common in the cyber world. There is a constant threat of data security to not just individuals or corporates but also government sectors. AI along with machine learning is used for the protection of data. It allows you to automate the detection of threat and combat without the involvement of humans. AI has been used for password protection and authenticity detection.

The IoT: AI is used to manage huge data flows and storage in the IoT network. With high-speed Internet networks and advanced sensors integrated into microcontrollers (MCUs), AI along with the IoT is creating a new wave of disruptive technologies.

With the explosion of the IoT, there are problems regarding data storage, delay, channel limitation and congestion in networks. One solution to solve these is to use AI in data mining, managing and controlling the congestion in networks. Techniques used in AI include fuzzy logic and neural networks in conjunction with the IoT network.

Finance and economics: Wall Street, financial district of the US, uses complex computer programs to do heavy jobs. These programs run on their own. A few years ago, stock markets plummeted, taking down a trillion dollars worth of market value, due to a malfunctioning ANI program.

Also, for example, when you deposit a cheque using your mobile banking app, it runs through a refined ANI system that can read the cheque much faster than humans. When you shop online, you are essentially feeding data into ANI systems.

Art and design: AI has been used to algorithmically generate objects that can be rendered digitally. It can generate new patterns with high speed, good efficiency and verisimilitude. Algorithm-driven design tools help you construct user interfaces, content and personalise user experiences.

Publishing tools such as Readymag and Squarespace have greatly simplified the work to the extent where you can get many high-quality templates and designs without having to pay for a designer. There are many other algorithm-driven design tools for graphic design including identity, drawing and illustrations.

AI solution providers offer tools and libraries to manipulate images and photos. Soon, AI will drive the next generation of apps for visual arts and creative designs.

An AI model, called CRAFT (Composition, Retrieval and Fusion Network), is being developed by researchers from University of Illinois, USA, and Allen Institute for Artificial Intelligence, USA. It can convert provided text descriptions into video clips of an animated series. To simplify, AI matches videos with word descriptions, builds a set of parameters and generates scenes.

Exploration and research: Defense Advanced Research Projects Agency (DARPA), USA, is working on its Artificial Intelligence Exploration (AIE) programme, which is a key component of the agency's broader AI programme.

AI in space exploration is gathering momentum, too. Over the next few years, new missions would be taken up by AI as we voyage to the Moon and the planets, and explore possibilities in space. AI is also being used in NASA's next Rover mission to planet Mars.

National Geographic Society and Microsoft are partnering to explore how AI can help us understand, engage and protect Earth.

AI is helping the oil and gas industry to preserve the ecosystem while discovering new resources.

As per a report by MarketsandMarkets, the global market for AI robots is expected to reach US$ 12.36 billion by 2023, growing at a CAGR of 28.78 per cent during the forecast period. Adoption of AI-powered robots for personal use, such as companionship and entertainment, support from governments worldwide to develop modern technologies, and financial assistance through government budgets or subsidies are some of the key factors driving the growth of this market.

AI-based service and industrial robots can learn repetitive tasks and even communicate with humans or, in some cases, with other peer robots. AI processors, network devices and AI platforms are the key components differentiating the latest AI robots from traditional ones.